

Hidden Markov Model for POS Tagging

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1 Introduction

This work uses the Wall Street Journal (WSJ) portion of the Penn Treebank dataset to develop a Part-of-Speech (POS) tagger for English based on a Hidden Markov Model (HMM) with Viterbi decoding. The goal is to decode the most likely tag sequence for unknown phrases and estimate transition and emission probabilities from trained data.

2 Dataset and Experimental Setup

We use the WSJ portion of the Penn Treebank corpus, split according to the standard convention:

- **Training set:** Sections 00–18
- **Development set:** Sections 19–21
- **Test set:** Sections 22–24

Each sentence consists of tokens in the format `word/TAG`. The dataset includes 36 core POS tags (45 when punctuation is included).

3 Baseline Model

3.1 Baseline Performance Summary (NLTK Tagger)

Metric	Value
Overall Accuracy	0.97
Weighted F1 Score	0.97
Macro F1 Score	0.86–0.87

Table 1: Baseline NLTK POS Tagger Performance

3.2 Observations

- High-frequency tags such as NN, DT, IN, and NNP achieve very high precision and recall.
- Rare tags (e.g., LS, PDT, RBS) suffer from undefined or zero precision due to extremely low support.

These results confirm that the evaluation pipeline and data splits are functioning correctly. This baseline will later be used to directly compare against the HMM-based POS tagger.

4 HMM Model

A bigram Hidden Markov Model (HMM) is implemented to perform POS tagging by modeling the joint probability of a word sequence and its corresponding tag sequence. The model estimates both transition and emission probabilities from the training data and decodes the most likely tag sequence using the Viterbi algorithm.

4.1 Transition Probabilities

Transition probabilities are defined as:

$$P(t_i | t_{i-1})$$

where t_i is the current tag and t_{i-1} is the previous tag. These probabilities are estimated from tag bigram counts in the training corpus.

4.2 Emission Probabilities

Emission probabilities model the likelihood of a word given a tag:

$$P(w_i | t_i)$$

where w_i is the observed word and t_i is the corresponding POS tag. These probabilities are estimated from word-tag co-occurrence counts.

4.3 Add- α Smoothing

Add- α smoothing is applied to both transition and emission probabilities to handle unseen events. The smoothing parameter α is tuned using the development set.

4.4 Decoding Algorithm

Decoding is performed using the Viterbi algorithm in log-space to avoid numerical underflow. The algorithm computes the most likely tag sequence by dynamic programming over the sentence length.

4.5 Special Tokens

The following special tokens are used:

- `<START>` — marks the beginning of a sentence
- `<STOP>` — marks the end of a sentence
- `<UNK>` — represents unseen words

5 Hyperparameter Tuning

The smoothing parameter α is tuned on the development set. The following values are evaluated:

α	Dev Accuracy
0.1	0.9425
0.25	0.9393
0.5	0.9345
1.0	0.9263
2.0	0.9139

Table 2: Development Set Accuracy for Different α Values

The best performance is achieved with $\alpha = 0.1$, which is used for the final HMM evaluation on the test set.